CHAPTER 9

Biomolecules

Analysis of Chemical Composition

1. Which of the following is a nucleotide? *(2024 Re)*

a. Uridine
b. Adenylic acid
c. Guanine
d. Guanosine

2. Which of the following are **not** fatty acids? *(2024 Re)*

A. Glutamic acid
B. Arachidonic acid
C. Palmitic acid
D. Lecithin
E. Aspartic acid

Choose the **correct** answer from the options given below:

a. C, D and E only
b. A and B only
c. A, D and E only
d. B and C only

3. Lecithin, a small molecular weight organic compound found in living tissues, is an example of: *(2024)*

a. Glycerides
b. Carbohydrates
c. Amino acids
d. Phospholipids

4. Given below are two statements: *(2022 Re)*

Statement-I: Amino acids have a property of ionizable nature of — NH_2 and — COOH groups, hence have different structures at different pH.

Statement-II: Amino acids can exist as Zwitter ionic form at acidic and basic pH.

In the light of the above statements, choose the most appropriate answer from the options given below:

a. Statement-I is incorrect but Statement-II is correct
b. Both Statement-I and Statement-II are correct
c. Both Statement-I and Statement-II are incorrect
d. Statement-I is correct but Statement-II is incorrect

5. Read the following statements and choose the set of correct statements *(2022)*

A. Lecithin found in the plasma membrane is a glycolipid
B. Saturated fatty acids possess one or more C=C bonds
C. Gingelly oil has lower melting point, hence remains as oil in winter
D. Lipids are generally insoluble in water but soluble in some organic solvents
E. When fatty acid is esterified with glycerol, monoglycerides are formed

Choose the correct answer from the options given below.

a. A, B and D only
b. A, B and C only
c. A, D and E only
d. C, D and E only

6. Following are the statements with reference to 'lipids'.

A. Lipids having only single bonds are called unsaturated fatty acids.
B. Lecithin is a phospholipid.
C. Trihydroxy propane is glycerol.
D. Palmitic acid has 20 carbon atoms including carboxyl carbon.
E. Arachidonic acid has 16 carbon atoms.

Choose the correct answer from the options given below. *(2021)*

a. C & D only
b. B & C only
c. B & E only
d. A & B only

7. Identify the basic amino acid from the following. *(2020)*

a. Glutamic acid
b. Lysine
c. Valine
d. Tyrosine

8. Identify the statement which is incorrect. *(2020-Covid)*

a. Glycine is an example of lipids
b. Lecithin contains phosphorus atom in its structure
c. Tyrosine possesses aromatic ring in its structure
d. Sulphur is an integral part of cysteine

9. A typical fat molecule is made up of: *(2016 - I)*

a. Three glycerol molecules and one fatty acid molecule
b. One glycerol and three fatty acid molecules
c. One glycerol and one fatty acid molecule
d. Three glycerol and three fatty acid molecules

10. A phosphoglyceride is always made up of: *(2013)*

a. A saturated or unsaturated fatty acid esterified to a phosphate group which is also attached to a glycerol molecule
b. Only a saturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached
c. Only an unsaturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached
d. A saturated or unsaturated fatty acid esterified to a glycerol molecule to which a phosphate group is also attached

Primary & Secondary Metabolites

11. Which of the following is not a secondary metabolite? *(2023-Manipur)*

a. Curcumin b. Morphine
c. Anthocyanin d. Lecithin

12. Match **List-I** with **List-II**. *(2023-Manipur)*

List-I		List-II	
(A)	Terpenoides	(I)	Codeine
(B)	Lectins	(II)	Diterpenes
(C)	Alkaloids	(III)	Ricin
(D)	Toxins	(IV)	Concanavalin A

Choose the **correct** answer from the options given below:

a. A-II, B-IV, C-III, D-I b. A-II, B-I, C-IV, D-III
c. A-II, B-III, C-I, D-IV d. A-II, B-IV, C-I, D-III

13. Match List-I with List-II: *(2022 Re)*

List-I		List-II	
(A)	Adenine	(I)	Pigment
(B)	Anthocyanin	(II)	Polysaccharide
(C)	Chitin	(III)	Alkaloid
(D)	Codeine	(IV)	Purine

Choose the correct answer from the options given below.

a. A-I, B-IV, C-III, D-II b. A-IV, B-I, C-II, D-III
c. A-IV, B-III, C-II, D-I d. A-III, B-I, C-IV, D-III

14. Identify the **incorrect** pair. *(2021)*

a. Toxin - Abrin b. Lectins - Concanavalin A
c. Drugs - Ricin d. Alkaloids - Codeine

15. Which of the following are **not** secondary metabolites in plants? *(2021)*

a. Amino acids, glucose b. Vinblastin, curcumin
c. Rubber, gums d. Morphine, codeine

16. Concanavalin A is *(2019)*

a. An alkaloid b. An essential oil
c. A lectin d. A pigment

Biomacromolecules, Proteins

17. Match **List-I** with **List-II**: *(2024 Re)*

List-I		List-II	
(A)	Primary structure of protein	(I)	Human haemoglobin
(B)	Secondary structure of protein	(II)	Disulphide bonds
(C)	Tertiary structure of protein	(III)	Polypeptide chain
(D)	Quaternary structure of protein	(IV)	α helix and β sheet

Choose the **correct** answer from the options given below:

a. A-III, B-IV, C-II, D-I b. A-III, B-II, C-I, D-IV
c. A-I, B-III, C-II, D-IV d. A-IV, B-III, C-II, D-I

18. Match List-I with List-II. *(2024)*

List-I		List-II	
(A)	GLUT-4	(I)	Hormone
(B)	Insulin	(II)	Enzyme
(C)	Trypsin	(III)	Intercellular ground substance
(D)	Collagen	(IV)	Enables glucose transport into cells

Choose the correct answer from the options given below:

a. A-(II) B-(III), C-(IV), D-(I)
b. A-(III), B-(IV), C-(I), D-(II)
c. A-(IV), B-(I), C-(II), D-(III)
d. A-(I), B-(II), C-(III), D-(IV)

19. Given below are two statements: *(2023)*

Statement-I: A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal)

Statement-II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type.)

In the light of the above statements, choose the correct answer from the options given below:

a. Statement-I is false but Statement-II is true.
b. Both Statement-I and Statement-II are true.
c. Both Statement-I and Statement-II are false.
d. Statement-I is true but Statement-II is false.

20. Primary proteins are also called as polypeptides because: *(2022 Re)*

a. They can assume many conformations
b. They are linear chains
c. They are polymers of peptide monomers
d. Successive amino acids are joined by peptide bonds

21. Match List-I with List-II. *(2022)*

List-I (Biological Molecules)		List-II (Biological function)	
(A)	Glycogen	(i)	Hormone
(B)	Globulin	(ii)	Biocatalyst
(C)	Steroids	(iii)	Antibody
(D)	Thrombin	(iv)	Storage product

Choose the correct answer from the options given below.

a. A-iv B-iii C-i D-ii
b. A-iii B-ii C-iv D-i
c. A-iv B-ii C-i D-iii
d. A-ii B-iv C-iii D-i

22. Which one of the following is the most abundant protein in the animals? *(2020)*

a. Collagen b. Lectin
c. Insulin d. Haemoglobin

23. Which of the following glucose transporters is insulin-dependent? *(2019)*

a. GLUT I b. GLUT II c. GLUT III d. GLUT IV

24. Which of the following are not polymeric? *(2017-Delhi)*
a. Nucleic acids b. Proteins
c. Polysaccharides d. Lipids

25. Which of the following is the least likely to be involved in stabilising the three–dimensional folding of most proteins? *(2016 - II)*
a. Hydrophobic interaction b. Ester bonds
c. Hydrogen bonds d. Electrostatic interaction

Polysaccharides, Nucleic Acids & Types of Bond

26. Match List-I with List-II. *(2024)*

List-I		List-II	
(A)	Lipase	(I)	Peptide bond
(B)	Nuclease	(II)	Ester bond
(C)	Protease	(III)	Glycosidic bond
(D)	Amylase	(IV)	Phosphodiester bond

Choose the correct answer from the options given below:
a. A-(II), B-(IV), C-(I), D-(III)
b. A-(IV), B-(I), C-(III), D-(II)
c. A-(IV), B-(II), C-(III), D-(I)
d. A-(III), B-(II), C-(I), D-(IV)

27. Cellulose does not form blue colour with Iodine because *(2023)*
a. It breaks down when iodine reacts with it.
b. It is a disaccharide.
c. It is a helical molecule.
d. It does not contain complex helices and hence cannot hold iodine molecules.

28. Match List-I with List-II. *(2023-Manipur)*

List-I		List-II	
(A)	Protein	(I)	C = C double bonds
(B)	Unsaturated fatty acid	(II)	Phosphodiester bond
(C)	Nucleic acid	(III)	Glycosidic bonds
(D)	Polysaccharide	(IV)	Peptide bonds

Choose the **correct** answer from the options given below:
a. A-II, B-I, C-IV, D-III b. A-IV, B-III, C-I, D-II
c. A-IV, B-I, C-II, D-III d. A-I, B-IV, C-III, D-II

29. Inulin is a polymer of: *(2023-Manipur)*
a. Fructose b. Galactose
c. Amino acids d. Glucose

30. Exoskeleton of arthropods is composed of: *(2022)*
a. Glucosamine b. Cutin
c. Cellulose d. Chitin

31. Match List-I with List-II *(2021)*

	List-I		List-II
A.	Protein	(i)	C = C double bonds
B.	Unsaturated fatty acid	(ii)	Phosphodiester bonds
C.	Nucleic acid	(iii)	Glycosidic bonds
D.	Polysaccharide	(iv)	Peptide bonds

Choose the correct answer from the options given below.

a.	A-i	B-iv	C-iii	D-ii
b.	A-ii	B-i	C-iv	D-iii
c.	A-iv	B-iii	C-i	D-ii
d.	A-iv	B-i	C-ii	D-iii

32. Identify the substances having glycosidic bond and peptide bond, respectively in their structure: *(2020)*
a. Glycerol, trypsin b. Cellulose, lecithin
c. Inulin, insulin d. Chitin, cholesterol

33. Match the following: *(2020-Covid)*

1.	Aquaporin	(i)	Amide
2.	Asparagine	(ii)	Polysaccharide
3.	Abscisic acid	(iii)	Polypeptide
4.	Chitin	(iv)	Carotenoids

Select the correct option:

	(1)	(2)	(3)	(4)
a.	(ii)	(iii)	(iv)	(i)
b.	(ii)	(i)	(iv)	(iii)
c.	(iii)	(i)	(ii)	(iv)
d.	(iii)	(i)	(iv)	(ii)

34. The two functional groups characteristic of sugars are: *(2018)*
a. Hydroxyl and methyl b. Carbonyl and methyl
c. Carbonyl and phosphate d. Carbonyl and hydroxyl

35. Which one of the following statements is wrong? *(2016 - I)*
a. Sucrose is a disaccharide
b. Cellulose is a polysaccharide
c. Uracil is a pyrimidine
d. Glycine is a sulphur containing amino acid

36. The chitinous exoskeleton of arthropods is formed by the polymerisation of: *(2015 Re)*
a. D-glucosamine
b. N-acetyl glucosamine
c. Lipoglycans
d. Keratin sulphate and chondroitin sulphate

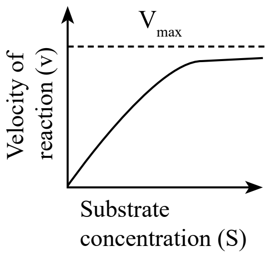
37. Which of the following biomolecules does have phosphodiester bond? *(2015 Re)*
a. Monosaccharides in polysaccharide
b. Amino acids in a polypeptide
c. Nucleic acids in a nucleotide
d. Fatty acids in a diglyceride

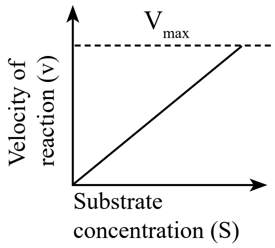
38. Which one of the following is a non-reducing carbohydrate? *(2014)*
a. Ribose 5-phosphate b. Maltose
c. Sucrose d. Lactose

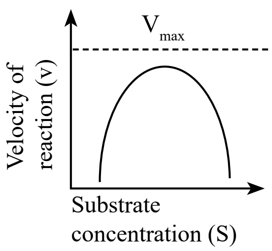
Enzymes

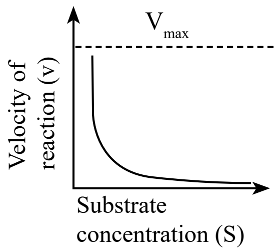
39. Ligases is a class of enzymes responsible for catalysing the linking together of two compounds. Which of the following bonds is **not** catalysed by it? *(2024 Re)*
a. C – C b. P – O c. C – O d. C – N

40. Which of the following graphs depicts the effect of substrate concentration on velocity of enzyme catalysed reaction? *(2024 Re)*

a.


b.


c.


d.


41. Enzymes that catalyse the removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds, are known as: *(2024 Re)*

a. Transferases
b. Oxidoreductases
c. Dehydrogenases
d. Lyases

42. Regarding catalytic cycle of an enzyme action select the correct sequential steps: *(2024)*

A. Substrate enzyme complex formation.
B. Free enzyme ready to bind with another substrate.
C. Release of products.
D. Chemical bonds of the substrate broken.
E. Substrate binding to active site.

Choose the correct answer from the options given below :

a. B, A, C, D, E
b. E, D, C, B, A
c. E, A, D, C, B
d. A, E, B, D, C

43. Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of: *(2024)*

a. Competitive inhibition
b. Enzyme activation
c. Cofactor inhibition
d. Feedback inhibition

44. The cofactor of the enzyme carboxypeptidase is *(2024)*

a. Flavin b. Haem c. Zinc d. Niacin

45. Malonate inhibits the growth of pathogenic bacteria by inhibiting the activity of *(2023)*

a. Dinitrogenase
b. Succinic dehydrogenase
c. Amylase
d. Lipase

46. Given below are two statements: *(2023)*

Statement-I: Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement-II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of the above statements, choose the correct answer from the options given below:

a. Statement-I is false but statement-II is true.
b. Both Statement-I and Statement-II are true.
c. Both Statement-I and Statement-II are false.
d. Statement-I is true but Statement-II is false.

47. In the enzyme which catalyses the breakdown of: *(2022 Re)*

$H_2O_2 \rightarrow H_2O + O_2$

the prosthetic group is:

a. Niacin
b. Nicotinamide adenine dinucleotide
c. Haem
d. Zinc

48. Match the following. Choose the correct option from the following: *(2020)*

	Column-I		Column-II
1.	Inhibitor of catalytic activity	(i)	Ricin
2.	Possess peptide bonds	(ii)	Malonate
3.	Cell wall material in fungi	(iii)	Chitin
4.	Secondary metabolite	(iv)	Collagen

	(1)	(2)	(3)	(4)
a.	(iii)	(i)	(iv)	(ii)
b.	(iii)	(iv)	(i)	(ii)
c.	(ii)	(iii)	(i)	(iv)
d.	(ii)	(iv)	(iii)	(i)

49. Consider the following statement :

A. Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.
B. A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.

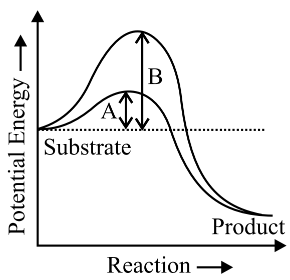
Select the correct option. *(2019)*

a. Both (A) and (B) are true.
b. (A) is true but (B) is false.
c. Both (A) and (B) are false.
d. (A) is false but (B) is true.

50. Which one of the following statements is correct, with reference to enzymes? *(2017-Delhi)*

a. Apoenzyme = Holoenzyme + Coenzyme
b. Holoenzyme = Apoenzyme + Coenzyme
c. Coenzyme = Apoenzyme + Holoenzyme
d. Holoenzyme = Coenzyme + Cofactor

51. Which of the following describes the given graph correctly? *(2016 - II)*



a. Endothermic reaction with energy A in absence of enzyme and B in presence of enzyme
b. Exothermic reaction with energy A in absence of enzyme and B in presence of enzyme
c. Endothermic reaction with energy A in presence of enzyme and B in absence of enzyme
d. Exothermic reaction with energy A in presence of enzyme and B in absence of enzyme.

52. A non-proteinaceous enzyme is: *(2016 - II)*

a. Ligase
b. Deoxyribonuclease
c. Lysozyme
d. Ribozyme

53. Which one of the following statements is incorrect? *(2015)*

a. The competitive inhibitor does not affect the rate of breakdown of the enzyme substrate complex
b. The presence of the competitive inhibitor decreases the K_m of the enzyme for the substrate
c. A competitive inhibitor reacts with the enzyme to form an enzyme inhibitor complex
d. In competitive inhibition, the inhibitor molecule is not chemically changed by the enzyme

54. Select the option which is not correct with respect to enzyme action: *(2014)*

a. Malonate is a competitive inhibitor of succinic dehydrogenase
b. Substrate binds with enzyme at its active site
c. Addition of lot of succinate does not reverse the inhibition of succinic dehydrogenase by malonate
d. A non-competitive inhibitor binds the enzyme at a site distinct from that which binds the substrate

55. The essential chemical components of many coenzymes are: *(2013)*

a. Vitamins
b. Proteins
c. Nucleic acids
d. Carbohydrates

56. Transition state structure of the substrate formed during an enzymatic reaction is: *(2013)*

a. Permanent and stable
b. Transient but stable
c. Permanent but unstable
d. Transient and unstable

Answer Key

1. (b)	**2.** (c)	**3.** (d)	**4.** (b)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (a)	**9.** (b)	**10.** (d)
11. (d)	**12.** (d)	**13.** (b)	**14.** (c)	**15.** (a)	**16.** (c)	**17.** (a)	**18.** (c)	**19.** (a)	**20.** (d)
21. (a)	**22.** (a)	**23.** (d)	**24.** (d)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (c)	**29.** (a)	**30.** (d)
31. (d)	**32.** (c)	**33.** (d)	**34.** (d)	**35.** (d)	**36.** (b)	**37.** (c)	**38.** (c)	**39.** (a)	**40.** (a)
41. (d)	**42.** (c)	**43.** (a)	**44.** (c)	**45.** (b)	**46.** (b)	**47.** (c)	**48.** (d)	**49.** (b)	**50.** (b)
51. (d)	**52.** (d)	**53.** (b)	**54.** (c)	**55.** (a)	**56.** (d)				